MILITARY SPECIFICATION

DECAL, LIGHT WEIGHT PAULIN

1. SCOPE

1.1 This specification covers one type of decal for use on life raft paulins to indicate the folding of the paulin for use as signal panels.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Specifications

Federal

NN-B-591	Boxes; Fiberboard, Wood-Cleated (For Domestic Shipment)
NN-B-621	Boxes, Wood, Nailed and Lock-Corner
NN-P-515	Plywood, Container Grade
TT-C-595	Colors; (For) Ready-Mixed Paints
LLL-B-631	Boxes; Fiber Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-676	Boxes, Set-Up, Paperboard

Military

JAN-P-106	Boxes, Wood, Nailed (Overseas Type)
JAN-P-108	Boxes, Fiberboard Corrugated and Solid (Military Type)
JAN-P-120	Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-B-4229	Boxes; Paperboard, Metal Edge
MIL-P-7249	Paulin, Light Weight

Standards

Military

MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Marking of U. S. Military Property

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Publications

Air Force-Navy Aeronautical Bulletin

143 Specifications and Standards; Use of

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specifications and Standards.- Specifications and standards for all materials, processes, parts, and equipment which are not specifically designated herein and which are necessary forthe execution of this specification shall be selected in accordance with Bulletin No. 143.

3.2 Materials.- The materials used in the manufacture of the decals shall be of high quality, suitable for the purpose, and shall conform to the applicable requirements specified herein. The decal film shall consist of an adhesion-backed film with a commercial decal paper backing applied over the adhesive, to be removed at time of use.

3.3 Design.- The design of the decal shall conform to Figure 1. The decal shall illustrate miniature paulins folded so as to convey the message indicated.

3.4 Film.- The film to be transferred shall be vinyl or any material that will meet the requirements specified herein.

3.4.1 Film Thickness.- The film of the decal after removal of the backing paper shall have a thickness of not less than 0.0015 inch nor more than 0.003 inch.

3.5 Backing Release.- The paper backing of the decal shall be removable in the "as received" condition, after immersion in water or other preparation.

3.6 Color.- The background of the transfer shall be Insignia White, Color No. 3715 of Specification TT-C-595. The miniature paulins shall be Insignia Blue, Color No. 3505 and Orange Yellow, Color No. 3305 of Specification TT-C-595 as required. All messages and "dotted" lines shall be Insignia Blue. All colors shall be an integral part of the film material. Messages shall be in condensed Gothic type.

3.7 Application.- The decal shall be suitable for application to a polymer or copolymer vinyl resin coated fabric such as is used in the construction of paulins conforming to Specification MIL-P-7249. The decal shall not break or stretch out of shape during application. The decals shall also be suitable for application to paulin fabrics which have been cleaned by washing with a 0.039 percent solution of alkyldimethylbenzylammonium chloride.

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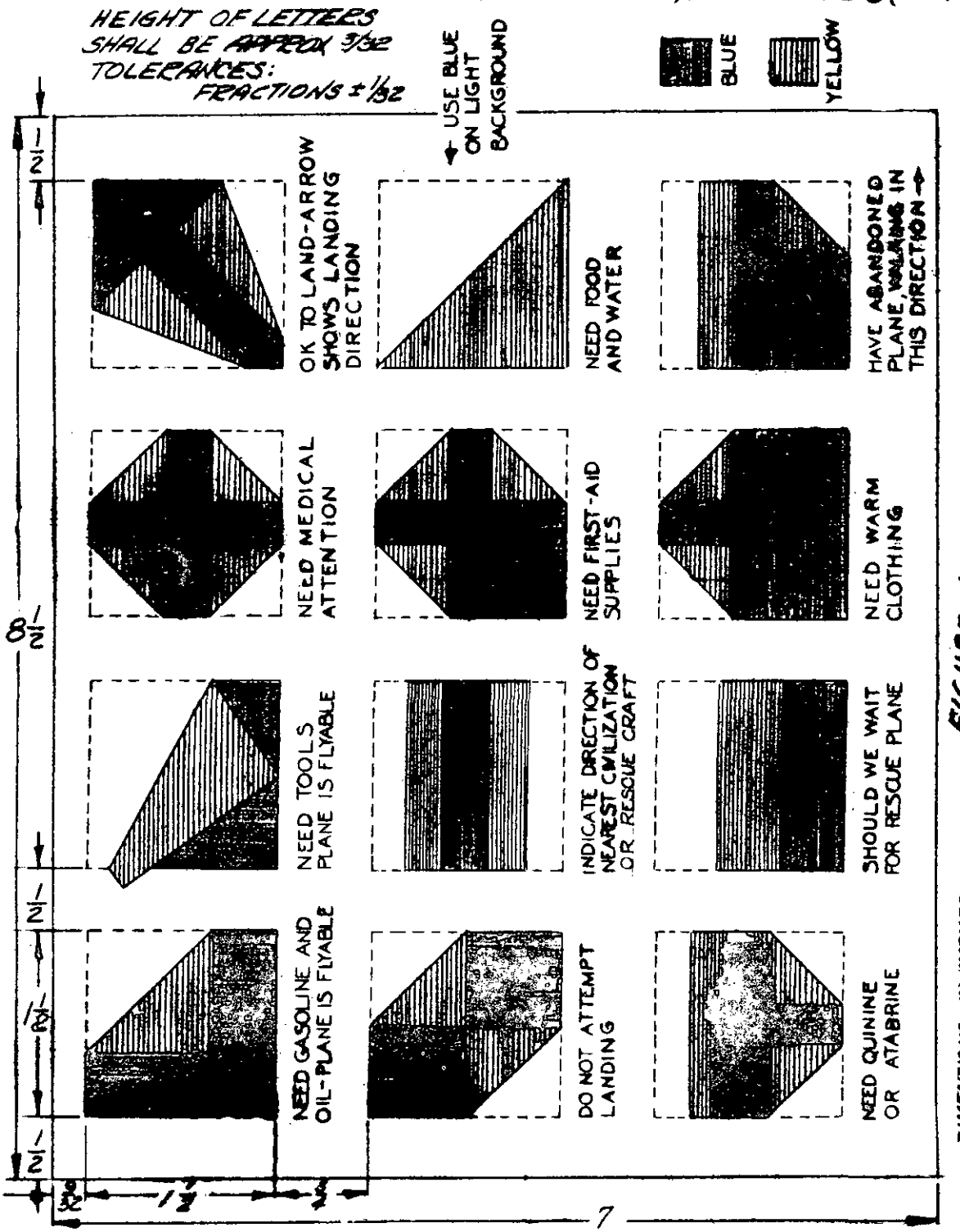


FIGURE 1

DIMENSIONS IN INCHES.

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3.7.1 Special solvents required for the application of the decal shall be considered a part of the decal and shall be furnished with the decal. The solvents required for the application of the decal shall be suitable for use at a temperature of plus 70 degrees plus or minus 20 degrees Fahrenheit and shall not be hazardous to using personnel.

3.8 Performance.-

3.8.1 Application and Adhesion Time.- The decal shall require not more than 60 seconds for application to the paulin material and not more than 60 minutes for adherence sufficient to permit the paulin to be handled normally.

3.8.2 Flexibility.-

3.8.2.1 Room Temperature.- The decal at plus 70 degrees Fahrenheit shall be flexible and shall bend sharply at a 90 degree angle.

3.8.2.2 Low Temperature.- The decal shall not crack or flake if bent sharply while being maintained at a temperature of minus 65 degrees Fahrenheit.

3.8.3 Heat Resistance.- The decal shall, when folded, withstand a temperature of plus 160 degrees Fahrenheit without evidence of adhering or peeling of the film.

3.8.4 Water Resistance.- The decal, when immersed in water, after application to the paulin, shall not loosen, flake, soften, or show any appreciable discolorattion.

3.8.5 Washing.- The decal shall not be visually affected when washed with a 0.039 percent solution of alkyldimethylbenzylammonium chloride.

3.9 Markings.-

3.9.1 Instructions.- Complete instructions for mounting the decal shall be printed on the upper surface of the backing paper. The backing shall be "keyed" to denote the top and the center of the sides of the decal.

3.9.2 Identification.- Each decal shall be marked on the upper surface of the backing paper in accordance with Standard MIL-STD-130 with the following information:

Decal, Light Weight Paulin
Specification MIL-D-4736

3.10 Workmanship.- The decal shall be constructed in a thoroughly workmanlike manner. The finished decal shall be clean and free from any defects that might affect appearance or functionability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection of the decals shall consist of the tests specified under Inspection Tests, subject to the test conditions specified herein.

4.2 Inspection Tests.-

4.2.1 Visual Inspection.- Each sample decal shall be visually inspected to determine conformance to the general design, coloring and marking requirements.

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4.2.2 Sampling Tests .- Two decals shall be selected for each test from each lot of 10,000 or fraction thereof on the order and subjected to the sampling tests. The Sampling tests shall be on decals that have met the requirements of the visual and dimensional inspection.

4.2.2.1 Examination of Product.- Each sample decal shall be examined to determine compliance with the dimension requirements of Figure 1.

4.2.2.2 Application and Adhesion Time.-

4.2.2.2.1 Application.- Each sample decal shall be tested to determine suitability for application to paulin fabric conforming to Specification MIL-P-7249 or equal. One of the samples shall be applied to paulin fabric that has been washed with a 0.039 percent solution of alkyldimethylbenzylammonium chloride. The time required for application shall be noted. Application time shall include time required for preparation of the decal for transfer, the transfer of the film, and the removal of the backing paper. The time for application shall not exceed 60 seconds.

4.2.2.2.2 Adhesion.- The decal, after an exposure to plus 70 degrees plus or minus 10 degrees Fahrenheit for 1 hour following the transfer to paulin fabric, shall adhere sufficiently to permit folding of decal without evidence of loosening or sticking. After 24 hours at the above temperature, the film will be examined for evidence of any separation from the paulin material. Any evidence of loosening or sticking shall be cause for rejection.

4.2.2.3 Flexibility.-

4.2.2.3.1 Room Temperature.- Each sample of paulin fabric with decal applied shall be pushed slowly over a horizontal surface having a 90 degree edge and the "bend" observed. The "bend" shall be across the decal. Any evidence of cracking or breaking of the decal shall be cause for rejection.

4.2.2.3.2 Low Temperature.- Each sample of paulin material with decal applied shall be exposed while on a flat smooth surface to a temperature of minus 65 degrees Fahrenheit for 1 hour. Without being removed from the temperature chamber, the paulin material shall be folded 180 degrees such that the fold falls across the decal. A 10-pound steel roller approximately 2 inches wide and at the same temperature of the test sample shall be pushed across the fold. The test sample shall be opened and folded again across the decal 90 degrees with respect to the first fold. The steel roller shall be pushed across the second fold. The decal shall be examined for cracking or flaking after removal from the test chamber. Any evidence of cracking or flaking shall be cause for rejection.

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4.2.2.4 Heat Resistance.- Each decal attached to paulin material shall be folded double, face to face, then back to back, and placed between two 4-1/2 by 4-1/2 by 1/8 inch glass plates such that the plates extend beyond the folds in the decal. A 4 pound weight shall be placed on the center of the top plate. While in this condition the decal shall be exposed to plus 160 degrees Fahrenheit for 24 hours. At the end of the exposure period, the sample shall be removed from the test chamber, taken from between the plates and allowed to cool for 1 hour. The sample shall then be unfolded slowly and examined for adhering or peeling of the film. Any evidence of adhering or peeling shall be cause for rejection.

4.2.2.5 Water Resistance.- Each sample shall be totally immersed in tap water at 70 degrees plus or minus 10 degrees Fahrenheit for 24 hours. At the end of the 24 hour period, the sample shall be removed from the water, permitted to dry, and the decal examined for peeling, blistering, or discoloration. Any evidence of peeling, blistering, or discoloration shall be cause for rejection.

4.2.2.6 Washing.- Each sample shall be placed on a flat surface and the transfer washed vigorously for 15 seconds with a ball of cotton or sponge thoroughly saturated with a 0.039 percent solution of alkyldimethylbenzylammonium chloride (Dis-N-Fex). After washing, the transfer shall be dried by means of blotting paper. Any evidence of peeling, blistering, or discoloration shall be cause for rejection.

4.2.3 Special Test.- One sample shall be taken, from each lot of 5000 or fraction thereof, on the order and tested for film thickness.

4.2.3.1 Film Thickness.- Two strips 1 by 7 inches shall be cut from the decal in such a manner that each color is represented. The backing paper and backing paper adhesive shall be removed with clear water and the films thoroughly dried. After drying, five micrometer readings shall be taken of the film at various spots with at least one reading for each color. The film shall have a thickness of not less than 0.0015 inch nor more than 0.003 inch.

5. PREPARATION FOR DELIVERY

5.1 General.- The requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 Interior Packaging.- Unless otherwise specified, decals shall be securely packaged within an interior container in accordance with Specification JAN-P-120, PPP-B-676, or MIL-B-4229. Each container shall be sealed with gummed tape in accordance with the requirements of Appendix 1 of Specification JAN-P-108. Each decal shall be separated by a wax paper that will not stick to or adversely affect the decals. The decals shall be stacked and packed flat. Each stack shall be wrapped in a suitable paper and sealed with gummed paper tape. The number of decals per container shall be dependent upon the ability of the decal to withstand shipment. Solvents shall be shipped separately in accordance with the requirements of the Interstate Commerce Commission. Liquid solvents shall be unit contained in commercial screw top cans having a capacity not exceeding 1 gallon.

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5.3 Exterior Packaging.-

5.3.1 Domestic Shipment.- Unless otherwise specified, the interior containers shall be overpacked in containers conforming to Specification NN-B-591, PPP-B-601, PPP-B-621, NN-B-621, LLL-B-631, or LLL-B-636, except that fiberboard having a minimum Mullen burst test of less than 275 pounds is prohibited. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3.2 Export Shipment.- Unless otherwise specified, for export shipment, the interior packages shall be packed in exterior shipping containers conforming to Specification PPP-B-601 or JAN-P-106. Plywood, if used, shall conform to Type 2, Class 2 of Specification NN-P-515. A waterproof caseliner conforming to Specification JAN-P-125 shall be provided and shall be sealed with adhesive conforming to Specification MIL-A-140.

5.4 Marking of Shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

Decal, Light Weight Paulin
Manufacturer's Part No. *
Specification MIL-D-4736

5.4.1 Solvents, if required, shall use the following nomenclature:

Solvent, Light Weight Paulin Decal
Manufacturer's Part No.& *
(Use with decals, MIL-D-4736)

*Applicable data to be entered by the contractor.

6. NOTES

6.1 Use.- Decals covered by this specification are intended for transfer to the lower left corner of the blue side of the No. 77 paulin, Specification MIL-P-7249, to provide instructions for folding the paulin for signalling.

6.2 Ordering Data.- Procurement documents and invitations for bids should specify whether domestic or export packing is required.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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